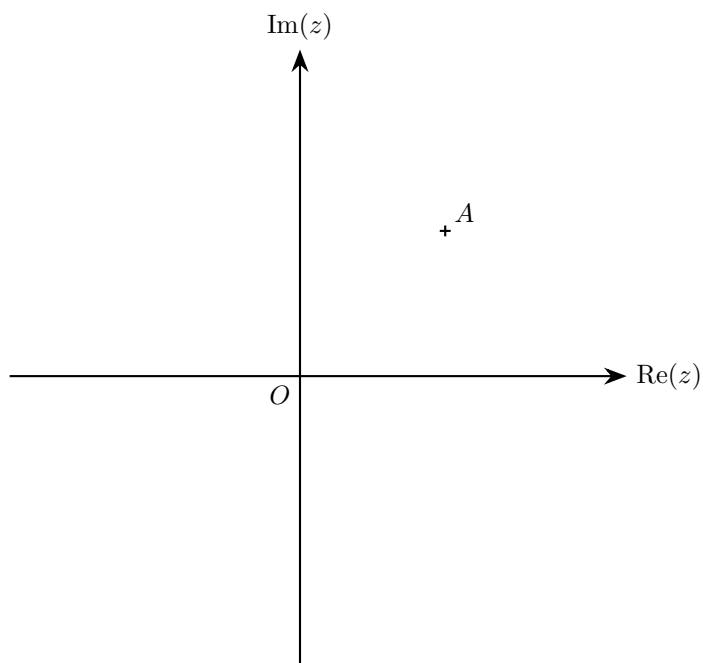


## Questions

|             |          |    |
|-------------|----------|----|
| Question 1  | 8 marks  | 2  |
| Question 2  | 10 marks | 3  |
| Question 3  | 13 marks | 4  |
| Question 4  | 10 marks | 5  |
| Question 5  | 5 marks  | 6  |
| Question 6  | 10 marks | 7  |
| Question 7  | 7 marks  | 8  |
| Question 8  | 12 marks | 9  |
| Question 9  | 9 marks  | 10 |
| Question 10 | 9 marks  | 11 |
| Question 11 | 14 marks | 12 |
| Question 12 | 8 marks  | 13 |
| Question 13 | 11 marks | 14 |



1. An Argand diagram with the point  $A$  representing a complex number  $z_1$  is shown in the diagram. The complex numbers  $z_2$ ,  $z_3$  and  $z_4$  are  $z_1 e^{\frac{1}{2}\pi i}$ ,  $z_1 e^{\pi i}$  and  $z_1 e^{\frac{3}{2}\pi i}$  respectively.
- (a) (i) On a copy of the Argand diagram, mark the points  $B$ ,  $C$  and  $D$  representing the complex numbers  $z_2$ ,  $z_3$  and  $z_4$ . [2]
- (ii) Show that  $z_1 + z_2 + z_3 + z_4 = 0$ . [2]
- (b) Given now that  $z_1$ ,  $z_2$ ,  $z_3$  and  $z_4$  are roots of the equation  $z^4 = -16$ , find these four roots, giving your answers in the form  $a + ib$ , where  $a$  and  $b$  are real and exact. [4]

- 2.** The five solutions of the equation  $z^5 = 16 + 16\sqrt{3}i$  are represented by points on an Argand diagram.

These points are the vertices of a regular pentagon  $P$  with centre at the origin.

- (a) Determine the five complex numbers that represent these vertices, giving your answers in the form  $re^{i\theta}$ , where  $0 \leq \theta < 2\pi$ . [4]
- (b) Sketch  $P$  on an Argand Diagram, taking care to label each vertex. [2]
- (c) Show that the area of  $P$  can be written  $k \sin \frac{2\pi}{5}$  where  $k$  is an integer to be determined. [4]

- 3.** (i) Find the 8th roots of  $-128 + 128\sqrt{3}i$  in the form  $re^{i\theta}$ , where  $r > 0$  and  $0 \leq \theta < 2\pi$ . [5]
- (ii) These 8th roots are represented in the Argand diagram, in order of increasing  $\theta$ , by the points  $A, B, C, D, E, F, G, H$ .  
Draw the Argand diagram, making clear which point is which. [2]
- (iii) The mid-point of  $BD$  is the point  $P$  which represents the complex number  $w$ .  
Find, in exact form, the modulus and argument of  $w$ . [3]
- (iv)  $w$  is an  $n$ th root of a real number  $a$ , where  $n$  is a positive integer. State the least possible value of  $n$  and find the corresponding value of  $a$ . [3]

4. Given that  $z_1 = 7 - i$  and  $z_2 = \frac{z_1}{1 - i}$ ,

(a) find  $z_2$  in the form  $a + bi$ , where  $a$  and  $b$  are real. [2]

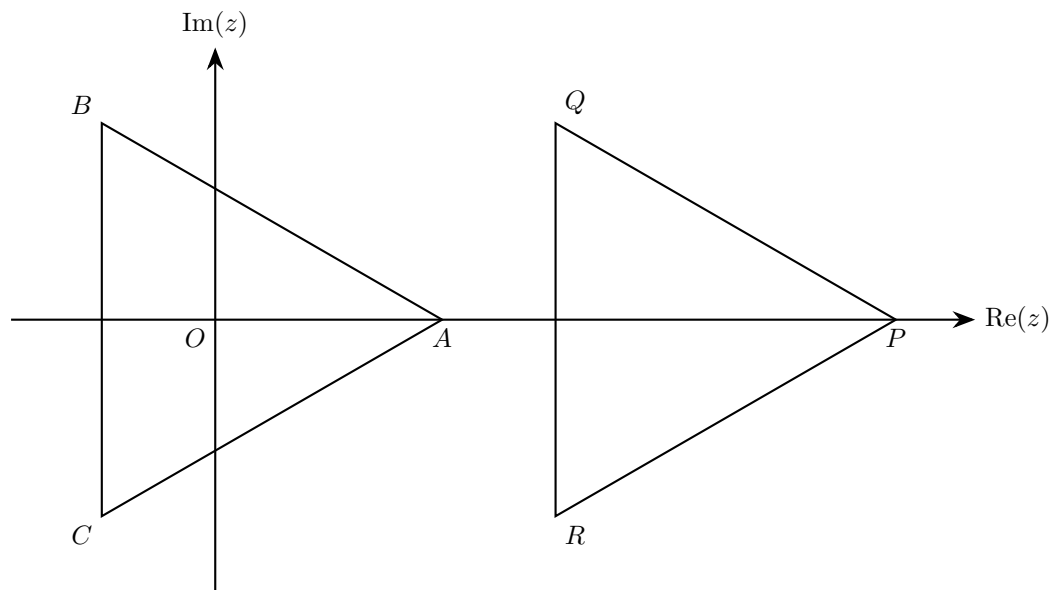
(b) Show on an Argand diagram the point  $P$  representing  $z_1$  and the point  $Q$  representing  $z_2$ . [2]

(c) Given that  $O$  is the origin, show that  $\angle OQP = \frac{\pi}{2}$ . [2]

The circle passing through the points  $O$ ,  $P$  and  $Q$  has centre  $C$ . Find

(d) the complex number represented by  $C$ . [2]

(e) the exact value of the radius of the circle. [2]



5. The cube roots of unity are represented on the Argand diagram above by the points  $A$ ,  $B$  and  $C$ .

The complex numbers represented by  $P$ ,  $Q$  and  $R$  are obtained by adding 2 to the complex numbers represented by  $A$ ,  $B$  and  $C$  respectively.

Determine a degree 6 polynomial equation with integer coefficients whose roots are the complex numbers represented by the points  $A$ ,  $B$ ,  $C$ ,  $P$ ,  $Q$  and  $R$ .

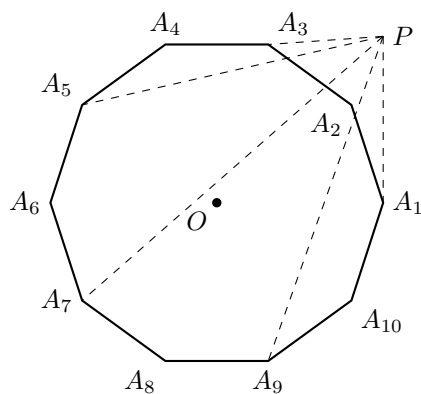
[5]

6. In an Argand diagram, the points  $A$ ,  $B$  and  $C$  are the vertices of an equilateral triangle with centre  $G$ . The point  $G$  represents the complex number  $1-2i$ , and the point  $A$  represents the complex number  $5+2i$ .

- (a) Find the complex numbers represented by the points  $B$  and  $C$ , giving your answers in the form  $x + iy$ , where  $x$  and  $y$  are real and exact. [6]

The points  $D$ ,  $E$  and  $F$  are the centroids of triangles  $GBC$ ,  $GCA$  and  $GAB$  respectively.

- (b) Find the exact area of triangle  $DEF$ . [4]



7. (a) Using the identity  $zz^* = |z|^2$ , or otherwise, show that if  $w$  is any root of unity then

$$|w - (1 + i)|^2 = 3 - (1 - i)w - (1 + i)w^* \quad [3]$$

The diagram shows a regular decagon  $A_1A_2 \dots A_{10}$  whose vertices all lie on the unit circle with centre at the origin  $O$  and  $A_1$  at  $(1, 0)$ . The point  $P$  lies in the same plane as the decagon and has coordinates  $(1, 1)$ .

- (b) Using the result from part (a), find

$$\sum_{r=1}^5 (PA_{2r-1})^2 \quad [4]$$

8. You are given the complex number  $\alpha = \cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5}$  and the equation  $z^5 = -1$ .

(a) Show that  $\alpha$  is a root of the equation. [2]

(b) Write down the other four roots of the equation. [1]

(c) Show that

$$\alpha^4 - \alpha^3 + \alpha^2 - \alpha + 1 = 0 \quad [2]$$

(d) Hence show that

$$\left(\alpha + \frac{1}{\alpha}\right)^2 - \left(\alpha + \frac{1}{\alpha}\right) - 1 = 0 \quad [3]$$

(e) Hence determine the value of  $\cos \frac{3\pi}{5}$  in the form  $a + b\sqrt{c}$ , where  $a$ ,  $b$  and  $c$  are rational numbers to be found. [4]

9. The points representing the complex numbers  $z_1 = 6 + 11i$  and  $z_2 = -8 - 3i$  are opposite vertices of a regular octagon,  $K$ , in the complex plane.  
The centre of  $K$  represents the complex number  $\alpha$ .

(a) Show that  $\alpha = -1 + 4i$ . [2]

Given that

$$\beta = \frac{1-i}{14}$$

(b) Show that  $\beta(z_1 - \alpha) = 1$  [2]

The vertices of  $K$  are given by the roots of the equation

$$(\beta(z - \alpha))^8 = 1$$

- (c) (i) Write down the roots of the equation  $w^8 = 1$  in the form  $re^{i\theta}$ . [1]  
(ii) Hence, or otherwise, determine the positions of the other six vertices of  $K$ , giving your answers as complex numbers in Cartesian form. [4]

- 10.** In an Argand diagram, the points  $A$  and  $B$  are represented by the complex numbers  $1 + 4i$  and  $7 + 2i$  respectively. The circle  $C$  passes through  $A$  and  $B$ , and its centre lies on the real axis.

**(a)** Find the equation of  $C$ , giving your answer in the form

$$|z - a| = b \quad a \in \mathbb{C}, b \in \mathbb{R} \quad [4]$$

The circle  $D$ , with equation  $|z - (1 + i)| = 5$ , intersects  $C$  at the points representing the complex numbers  $z_1$  and  $z_2$ .

**(b)** Find the complex numbers  $z_1$  and  $z_2$ . [5]

11. The cubic equation

$$z^3 + pz^2 + qz + 24 = 0$$

where  $p$  and  $q$  are real, has roots  $\alpha$ ,  $\beta$  and  $\gamma$ .

It is given that  $\alpha = -2 + 2i$ .

- (a) (i) Write down another root,  $\beta$ , of the equation. [1]  
(ii) Find the third root,  $\gamma$ . [3]  
(iii) Find the values of  $p$  and  $q$ . [3]

- (b) (i) Express  $\alpha$  in the form  $re^{i\theta}$ , where  $r > 0$  and  $-\pi < \theta \leq \pi$ . [2]  
(ii) Hence show that

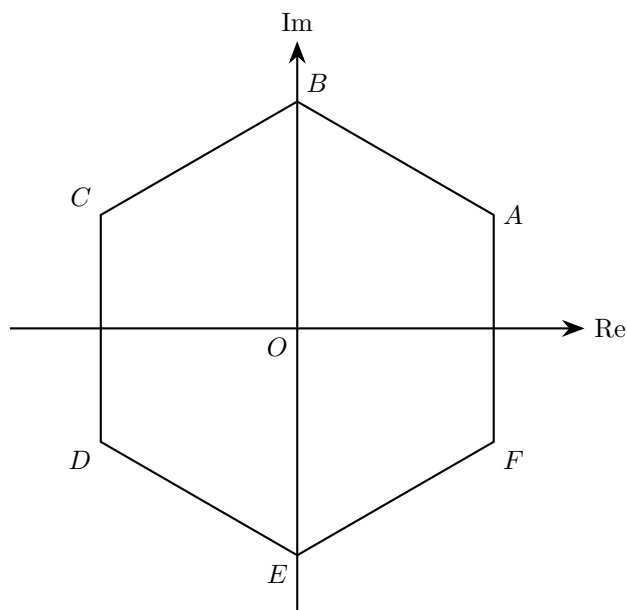
$$(-2 + 2i)^n = (2\sqrt{2})^n \left( \cos \frac{3n\pi}{4} + i \sin \frac{3n\pi}{4} \right) \quad [2]$$

- (iii) Hence show that

$$\alpha^n + \beta^n + \gamma^n = 2(2\sqrt{2})^n \cos \frac{3n\pi}{4} + (-3)^n$$

where  $n$  is an integer.

[3]



**12.** In the diagram, the points  $A$ ,  $B$ ,  $C$ ,  $D$ ,  $E$  and  $F$  represent the complex sixth roots of  $-729$  on an Argand diagram. The centroids of triangles  $OAB$ ,  $OBC$ ,  $OCD$ ,  $ODE$ ,  $OEF$  and  $OFA$  are  $G$ ,  $H$ ,  $I$ ,  $J$ ,  $K$  and  $L$  respectively.

- (a) Write down, in exponential form, the complex numbers represented by the points  $A$ ,  $B$ ,  $C$ ,  $D$ ,  $E$  and  $F$ . [2]
- (b) When these complex numbers are multiplied by the complex number  $w$ , the resulting complex numbers are represented by the points  $G$ ,  $H$ ,  $I$ ,  $J$ ,  $K$  and  $L$ . Find  $w$  in exponential form. [4]
- (c) You are given that  $G$ ,  $H$ ,  $I$ ,  $J$ ,  $K$  and  $L$  represent roots of the equation  $z^6 = p$ . Find  $p$ . [2]

13. (i) Write down, in any form, all the complex roots of the equation

$$w^{11} = -1 \quad [2]$$

- (ii) Explain why the equation

$$(z+1)^{11} = -(z-1)^{11}$$

has exactly 10 non-real roots and show that they may be expressed in the form

$$-i \cot\left(\frac{m\pi}{22}\right) \quad [6]$$

where  $m = \pm 1, \pm 3, \pm 5, \pm 7, \pm 9$ .

- (iii) Show that

$$\left\{-i \cot\left(\frac{m\pi}{22}\right)\right\} \left\{i \cot\left(\frac{m\pi}{22}\right)\right\} = \cot^2\left(\frac{m\pi}{22}\right) \quad [1]$$

- (iv) Given that the product of the non-zero roots of the equation in part (ii) is 11, find the value of

$$\tan^2\left(\frac{\pi}{22}\right) \tan^2\left(\frac{3\pi}{22}\right) \tan^2\left(\frac{5\pi}{22}\right) \tan^2\left(\frac{7\pi}{22}\right) \tan^2\left(\frac{9\pi}{22}\right) \quad [2]$$